MODULES 6 & 7 | EXPANDED NCLEX / HESI REVIEW

Comprehensive printable edition | Black and white | Landscape

Nine assigned chapters, organized by module and chapter. This edition expands pathophysiology, assessment, warning signs, diagnostics, laboratory values, medications, nursing interventions, patient teaching, and priority-setting. It is a structured study synthesis of the supplied chapter materials, not a reproduction of every textbook paragraph. Refer to your assigned textbook for complete tables and institutional protocols.

Contents

• MODULE 6: Chapter 19 Reproduction; Chapter 20 Sexuality; Women's Reproductive Health and Gynecologic Disorders; Infertility, Assisted Reproductive Technologies and Contraception

• MODULE 7: Chapter 8 Nursing Care of the Family During Pregnancy; Chapter 9 Maternal Nutrition; Chapter 10 Assessment of High-Risk Pregnancy; Chapter 11 High-Risk Pregnancy: Preexisting Health Conditions; Chapter 12 High-Risk Pregnancy: Gestational Conditions

• Appendix: Rapid comparison charts, priority cues and practice questions (answers withheld)

How to use this guide

• Study one chapter at a time. Explain why each manifestation occurs, not just what it is.

• For every medication, identify the purpose, what to monitor, and the adverse effect that changes nursing action.

• For every abnormal finding, ask: stable or unstable? Maternal airway/circulation threatened? Fetal oxygenation threatened?

• Practice questions at the end deliberately omit answers so you can test yourself.

MODULE 6 | CHAPTER 19 - REPRODUCTION (GIDDENS)

1. Core physiology: from gametes to pregnancy

Oogenesis begins before birth; immature oocytes remain arrested until hormonal maturation. Usually one dominant follicle releases an ovum each cycle. **Spermatogenesis** occurs continuously after puberty in the testes; sperm quality depends on hormonal, anatomic, and environmental factors.

Menstrual phase	Hormonal pattern	Clinical significance
Follicular/proliferative	FSH recruits follicles; estrogen rises	Endometrium thickens; variable cycle length is largely due to follicular phase.
Ovulation	LH surge triggers follicular rupture	Highest conception likelihood in days leading up to and near ovulation.
Luteal/secretory	Corpus luteum secretes progesterone	Endometrium becomes receptive to implantation.
Menstruation	Progesterone/estrogen fall if no pregnancy	Endometrial lining sheds.

Fertilization most commonly occurs in the ampulla of the fallopian tube. The blastocyst travels to the uterus and implants about 6-10 days after fertilization. Trophoblast contributes to placental formation; embryoblast forms the embryo. hCG supports corpus luteum progesterone secretion early in pregnancy.

Embryonic period: Major organ formation occurs during approximately weeks 3-8 after fertilization; exposure to teratogens can cause structural malformations. **Fetal period:** Beginning around week 9 after fertilization, growth and functional maturation predominate. Obstetric gestational age is usually counted from the last menstrual period, about two weeks earlier than fertilization age.

2. Pregnancy history and terminology

Gravida = number of pregnancies including current; **para** = number of pregnancies reaching the defined viable/birth gestation threshold, not number of infants. **GTPAL** = total pregnancies, term deliveries, preterm deliveries, abortions/losses, living children. A multifetal delivery counts as one T or P event but multiple living children.

Example: Pregnant now; previously delivered one term infant and preterm twins; one miscarriage. G4 T1 P1 A1 L3.

3. Infertility, pregnancy loss, and unintended pregnancy

• Infertility evaluation generally begins after 12 months of unsuccessful intercourse under age 35 or 6 months at 35 and older; evaluate both partners.

• Female causes include anovulation, hormonal dysfunction, tubal obstruction, uterine pathology, endometriosis, nutritional or systemic disease. Male causes include low sperm count/motility, varicocele, obstruction, hormonal disorders, infections, heat/toxin exposure.

• Early pregnancy loss commonly involves bleeding/cramping; evaluate hemodynamic stability, ultrasound and serial hCG as ordered, Rh status, pain, infection and emotional needs.

• Ectopic pregnancy: implantation outside uterine cavity, often tubal. Severe unilateral pain, shoulder pain, syncope, tachycardia or hypotension may signal rupture and internal bleeding.

• Offer nonjudgmental contraceptive counseling and support for all pregnancy outcomes; assess coercion, access to care and patient goals.

4. Prenatal risk screening and nursing role

Risk domain	Examples	Nursing response
Biophysical	Chronic disease, age, multifetal gestation, previous complications	Obtain history, baseline assessment, indicated specialist referral.
Psychosocial	Mental health, substance use, violence, support limitations	Private screening, trauma-informed care, safety resources.
Sociodemographic	Limited prenatal access, transportation, food insecurity	Identify barriers and coordinate resources.
Environmental	Mercury, lead, radiation, smoking, medications	Review exposures and teach risk reduction.

NCLEX PRIORITY: Bleeding plus severe abdominal pain or shock in early pregnancy requires urgent escalation. Maternal stabilization is the first priority.

MODULE 6 | CHAPTER 20 - SEXUALITY (GIDDENS)

1. Sexuality as a nursing concept

Sexuality includes sex, gender identity and expression, sexual orientation, intimacy, pleasure, relationships and reproduction. It changes across the lifespan and is influenced by health, medications, culture, psychological factors and experiences. Do not assume identity, anatomy, sexual practices or partner gender.

Sexual response: Desire, arousal and orgasm may be influenced by neurologic, vascular, hormonal and psychological mechanisms. A patient may have distressing changes in desire, arousal, pain or orgasm due to illness, medications, menopause, pelvic conditions or relationship factors.

2. Sexual history: the CDC 5 Ps

P	Sample clinical focus	Nursing principle
Partners	Number/gender of partners; relevant exposures	Use neutral, nonjudgmental questions.
Practices	Types of sexual contact	Guides site-specific testing.
Protection	Condom/barrier use and prevention	Offer individualized risk reduction.
Past STIs	Previous infections, testing and treatment	Assess reinfection risk and follow-up.
Pregnancy intention	Whether/when pregnancy is desired	Offer contraception or preconception care.

3. Infections, prevention and safety

• STIs are frequently asymptomatic. Obtain testing based on exposure, anatomy, pregnancy status, age and guidelines.

• Untreated infections can lead to pelvic inflammatory disease, infertility, pregnancy complications and neonatal infection.

• HPV vaccination prevents many HPV-related cancers and genital warts; it does not treat an existing infection.

• Condoms reduce but do not eliminate STI transmission; hormonal contraception and IUDs do not prevent STIs.

• Ask about intimate partner violence privately, assess immediate safety, document objectively, and offer appropriate resources.

4. Sexual dysfunction and lifespan changes

Issue	Possible causes	Nursing focus
Dyspareunia	Vaginal dryness, infection, endometriosis, pelvic floor dysfunction	Assess location, onset, infection and trauma history; avoid minimizing pain.
Erectile dysfunction	Vascular disease, diabetes, medications, psychological factors	Screen cardiometabolic contributors; medication education.
Menopause symptoms	Declining ovarian estrogen; vasomotor and genitourinary changes	Discuss symptom treatment, bone health and individualized risks.

Issue	Possible causes	Nursing focus
Medication effects	Antidepressants, antihypertensives and other drugs	Review medications without advising abrupt discontinuation.

NCLEX PRIORITY: A report of sexual assault or coercion requires a supportive, nonjudgmental response, attention to immediate safety, consent and appropriate forensic/medical options.

MODULE 6 | WOMEN'S REPRODUCTIVE HEALTH & GYNECOLOGIC DISORDERS

1. Menstrual abnormalities: definitions and reasoning

Condition	Typical cues	Workup and care
Primary dysmenorrhea	Cramping with menses; prostaglandin-mediated contractions	NSAIDs often effective; heat and symptom tracking.
Secondary dysmenorrhea	New/worsening pain, sometimes dyspareunia or heavy bleeding	Investigate endometriosis, fibroids, adenomyosis or infection.
Amenorrhea	Absent periods	Rule out pregnancy; evaluate weight, exercise, stress, thyroid, prolactin and ovarian function.
PMS / PMDD	Cyclic physical and mood symptoms; PMDD causes marked impairment	Symptom diary, supportive care, prescribed hormonal or mood treatments.
Abnormal uterine bleeding	Excessive, prolonged or irregular bleeding	Pregnancy test, CBC and hemodynamic assessment; evaluate structural/hormonal causes.

2. Endometriosis and uterine fibroids

Endometriosis: Endometrial-like tissue outside uterus leads to inflammation, adhesions and pain. Typical clues include cyclic pelvic pain, painful intercourse, painful defecation during menses and infertility. Diagnosis may involve imaging and/or laparoscopy. Treatments can include NSAIDs, hormonal suppression and surgery. Teach chronic-pain management and fertility implications.

Leiomyomas (fibroids): Benign uterine smooth-muscle tumors can cause heavy menstrual bleeding, pelvic pressure, urinary frequency and anemia. Ultrasound helps evaluate. Management depends on symptoms, fertility goals and size/location; options include medications and procedures. Assess fatigue, pallor and blood loss.

3. Vaginal and sexually transmitted infections

Infection	Common findings	High-yield teaching
Chlamydia	Often asymptomatic; cervicitis or pelvic pain	Treat patient/partners as indicated; abstain until treatment complete.
Gonorrhea	Discharge, dysuria or asymptomatic	Appropriate antibiotic therapy; partner care; assess co-infection.
Trichomoniasis	Vaginal irritation, discharge; may be asymptomatic	Prescribed antimicrobial therapy and partner management.
Bacterial vaginosis	Thin discharge, fishy odor	Microbiome imbalance, not classified as a typical STI.
Candidiasis	Pruritus, thick discharge, erythema	Antifungal therapy; consider diabetes/recurrent symptoms.
Genital herpes	Painful vesicles/ulcers; recurrent episodes	Antivirals reduce symptoms/shedding but do not eradicate HSV.
HPV	Often no symptoms; may cause warts/dysplasia	Vaccination and recommended cervical screening.

4. Pelvic inflammatory disease (PID)

Ascending upper reproductive tract infection can cause endometritis, salpingitis and pelvic inflammation. Findings may include lower abdominal pain, cervical motion tenderness, uterine/adnexal tenderness, fever and abnormal discharge. Complications include infertility, ectopic pregnancy and chronic pelvic pain. Nursing actions: prompt antibiotics as ordered, assess pregnancy and severity, teach partner treatment and follow-up, recognize signs requiring hospitalization.

5. Breast concerns, menopause and malignancy awareness

• Benign breast changes may be cyclic, cystic or fibroadenomatous; new masses, nipple discharge or skin changes still need clinical evaluation.

• Breast cancer warning signs may include persistent mass, skin dimpling, nipple retraction or abnormal unilateral discharge. Avoid assuming a painless mass is benign.

• Perimenopause can involve irregular cycles and vasomotor symptoms. Postmenopausal bleeding is abnormal and requires evaluation for endometrial pathology.

• Assess for anemia with heavy bleeding and for psychosocial effects of reproductive illness, surgery or cancer.

NCLEX PRIORITY: Unstable vaginal bleeding (tachycardia, hypotension, syncope, pallor, altered mentation) requires circulation-focused emergency assessment before routine teaching.

MODULE 6 | INFERTILITY, ART & CONTRACEPTION

1. Causes and diagnostic workup

Fertility requires ovulation, patent reproductive tract, adequate sperm and implantation. Both partners should be assessed. Consider ovulatory dysfunction (including PCOS), tubal disease from PID, endometriosis, uterine abnormalities, semen factors, endocrine disorders, age and environmental factors.

Assessment	What it evaluates	Teaching / precaution
Semen analysis	Count, morphology, motility and volume	Follow laboratory collection/abstinence instructions.
Ovulation predictor / progesterone	LH surge or evidence of ovulation	Timing matters; anovulation may require endocrine workup.
FSH, LH, prolactin, TSH and other labs	Pituitary, ovarian and thyroid function	Results depend on cycle timing and clinical context.
HSG	Uterine cavity and tubal patency	Assess pregnancy possibility, infection and contrast allergy.
Transvaginal ultrasound	Follicles, uterus and adnexa	Often used for monitoring stimulation.
Laparoscopy	Direct visualization of endometriosis/adhesions	Postprocedure monitoring and recovery teaching.

2. Fertility medications

Drug class / example	Why used	Major nursing monitoring
Clomiphene citrate	Stimulates ovulation through hypothalamic/pituitary effects	Multiple gestation, visual symptoms, ovarian enlargement.
Letrozole	Aromatase inhibitor used for ovulation induction	Pregnancy exclusion before cycle, response to treatment.
Gonadotropins (FSH/LH)	Directly stimulate follicle development	Serial ultrasound/labs; ovarian hyperstimulation and multifetal risk.
hCG trigger	Triggers final follicular maturation/ovulation	Timing of intercourse, IUI or retrieval is critical.
Progesterone support	Supports endometrial environment in selected ART cycles	Correct route/timing and adherence.
Metformin	May address insulin resistance in selected patients	GI effects and individualized metabolic monitoring.

3. ART techniques and complications

- **IUI:** Prepared sperm placed in uterus near ovulation; fertilization occurs in the body.
- **IVF:** Ovarian stimulation, egg retrieval, fertilization in lab, embryo culture and transfer.
- **ICSI:** Single sperm injected into an oocyte; may be used with significant male-factor infertility.

• **Ovarian hyperstimulation syndrome:** Abdominal distension/pain, rapid weight gain, nausea, reduced urine output or dyspnea after stimulation. Escalate because fluid shifts and thromboembolism may occur.

• Psychosocial care: acknowledge grief, cost, ethical choices, repeated cycles, donor issues and variable outcomes without promising pregnancy.

4. Contraception: compare mechanisms and teaching

Method	How it works	Key considerations
Combined estrogen-progestin pill/patch/ring	Suppresses ovulation	Assess estrogen contraindications, VTE risks, adherence and missed-dose plan.
Progestin-only pill	Primarily thickens cervical mucus; some suppress ovulation	Timing/adherence important; irregular bleeding common.
Implant	Long-acting progestin	Highly effective; irregular bleeding possible.
Hormonal IUD	Thickens cervical mucus, alters endometrium	Long-acting; often decreases menstrual bleeding.
Copper IUD	Copper impairs sperm function	Hormone-free; may increase cramps/bleeding.
Condom	Physical barrier	STI risk reduction; correct use every encounter.
Sterilization	Permanent interruption of reproductive pathway	Counsel permanence and alternatives.
Emergency contraception	Prevents pregnancy after unprotected intercourse	Time-sensitive; method choice depends on timing and circumstances.

NCLEX PRIORITY: Combined estrogen-containing contraception requires individualized contraindication assessment, including migraine with aura, thrombosis history and certain cardiovascular risks.

MODULE 7 | CHAPTER 8 - NURSING CARE OF FAMILY DURING PREGNANCY

1. Confirming and dating pregnancy

Evidence category	Examples	Interpretation
Presumptive	Amenorrhea, fatigue, nausea, breast tenderness, quickening	Experienced/reported; not diagnostic.
Probable	Positive urine/serum hCG, Goodell, Chadwick and Hegar signs	Strongly suggests but does not independently prove viable pregnancy.
Positive	Fetal cardiac activity/visualization on ultrasound; fetal movement palpated by examiner	Confirms pregnancy.

Goodell: cervical softening. **Chadwick:** bluish vaginal/cervical coloration. **Hegar:** soft lower uterine segment. **Naegele rule:** first day of LMP minus three months plus seven days and one year, based on a regular 28-day cycle. Early ultrasound is often most accurate for dating.

2. Initial prenatal history and routine assessment

• Obtain GTPAL, prior pregnancy complications, gestational age, chronic illness, surgery, medication and allergy history, genetics and family history.

• Ask privately about depression, anxiety, IPV, alcohol/tobacco/drug exposure and support/resources.

• Assess BP, weight/BMI, edema when clinically relevant, urine symptoms, nutrition and fetal growth.

• Fundal height in centimeters after approximately 20 weeks often approximates weeks of gestation (allow clinical variation).

• Typical fetal baseline heart rate: 110-160/min. Decreased fetal movement needs assessment.

• Supine hypotensive syndrome: gravid uterus compresses vena cava; reposition laterally if dizzy or hypotensive while supine.

3. Screening schedule and why it matters

Test / visit focus	Usual timing or reason	Nursing significance
CBC, ABO/Rh, antibody screen	Initial visit; repeats as indicated	Anemia and alloimmunization risk.
Rubella/varicella immunity	Initial assessment	Live vaccines deferred until after pregnancy when indicated.
HIV, syphilis, hepatitis B and STI tests	Initial; repeat per risk and protocol	Prevention of maternal-neonatal transmission.
Urine testing/culture	Early pregnancy and when symptomatic	Asymptomatic bacteriuria can progress to pyelonephritis.
Anatomy ultrasound	Typically mid-second trimester	Structural development and placental location.
Gestational diabetes screen	Usually 24-28 weeks	Identify glucose intolerance.
GBS rectovaginal screening	Typically 36-37 weeks	Intrapartum antibiotics if indicated.

Test / visit focus	Usual timing or reason	Nursing significance
Rh immune globulin	For eligible unsensitized Rh-negative patients	Commonly at about 28 weeks and after qualifying sensitizing events.

4. Patient teaching: expected versus dangerous

Common/expected discomfort	Warning that changes action
Mild nausea, fatigue, frequency	Inability to retain fluids, dehydration, fever or dysuria.
Mild dependent swelling	Severe headache, visual symptoms, RUQ pain or sudden dyspnea.
Musculoskeletal back pain	Regular contractions, bleeding or fluid leakage.
Normal leukorrhea	Foul odor, itching, pain or sudden gush of fluid.

Teach healthy activity as appropriate, food safety, prenatal supplements, dental care, substance avoidance, recommended vaccines and signs of preterm labor. Include partner/family if desired by the patient. Use teach-back and address literacy, cultural needs and access barriers.

NCLEX PRIORITY: Pregnant patients with severe headache, visual changes, vaginal bleeding, leakage of fluid, severe abdominal pain or decreased fetal movement should not be reassured without assessment.

MODULE 7 | CHAPTER 9 - MATERNAL NUTRITION

1. Physiologic nutrition demands

Pregnancy increases demands for energy, protein, iron, folate and other micronutrients to support placenta, maternal blood volume and fetal growth. Nutritional assessment should include prepregnancy BMI, weight trend, dietary recall, supplements, food security, nausea/vomiting, cultural patterns and possible eating disorders.

Prepregnancy BMI	Recommended singleton gain	Rationale
Below 18.5	28-40 lb	Reduce inadequate fetal growth risk.
18.5-24.9	25-35 lb	Supports expected maternal and fetal growth.
25-29.9	15-25 lb	Avoid excessive gain and related complications.
30 or above	11-20 lb	Individualize counseling and avoid weight stigma.

2. Nutrient-specific high-yield details

Nutrient	Why important	Practical teaching
Folic acid	Neural tube development, especially before/early pregnancy	Take prescribed prenatal folate; fortified grains and greens support intake.
Iron	Maternal RBC mass, fetal iron stores, anemia prevention	Take with vitamin C if tolerated; separate from calcium/antacids when possible; constipation/dark stool may occur.
Calcium + vitamin D	Fetal skeletal mineralization	Dairy/fortified alternatives; supplement as indicated.
Protein	Maternal tissues, placenta and fetal growth	Include varied protein sources across meals.
Iodine	Maternal thyroid and fetal brain development	Iodized salt, dairy and seafood as appropriate.
Vitamin B12	Neurologic and RBC function	Assess vegan diets or malabsorption; supplement as indicated.
Choline	Fetal brain development	Eggs, meat, legumes and other sources.

3. Food safety, pica and special circumstances

• Avoid alcohol; no safe amount has been established for pregnancy.

• Avoid unpasteurized dairy/juices, raw or undercooked animal foods, and high-mercury fish. Use appropriate food handling and refrigeration.

• Choose low-mercury fish as recommended; fish can provide useful omega-3 nutrients.

• Caffeine should be limited according to obstetric guidance; review energy drinks and herbal products.

• Pica (ice, clay, dirt, starch) can accompany iron deficiency or create toxicity/exposure risk; ask without judgment.

• Severe vomiting, poor weight gain, multifetal pregnancy, food insecurity, diabetes, bariatric surgery or dietary restrictions may warrant dietitian referral.

4. Interpreting nutritional assessment

Iron deficiency clues: Fatigue, pallor, low hemoglobin/hematocrit, low ferritin, pica. **Hyperemesis clues:** Weight loss, dehydration, ketonuria in some cases and electrolyte abnormalities. **Priority:** Evaluate maternal hydration and fetal growth rather than focusing only on the number on the scale.

MODULE 7 | CHAPTER 10 - ASSESSMENT OF HIGH-RISK PREGNANCY

1. Why antepartum surveillance is performed

High-risk assessment identifies fetuses at risk from interrupted oxygenation while avoiding unnecessary intervention when fetal status is reassuring. Indications include maternal diabetes, hypertension, renal or autoimmune disease, fetal growth restriction, oligohydramnios, decreased movement and prior stillbirth.

2. Nonstress test (NST)

NST measures fetal heart-rate accelerations associated with fetal movement. A reactive NST at 32 weeks or later usually requires at least two accelerations of 15 bpm above baseline lasting 15 seconds within 20 minutes. Before 32 weeks, 10-by-10 criteria may be used. A nonreactive result may reflect sleep cycle, prematurity or compromised oxygenation; it warrants additional assessment, not automatic diagnosis of fetal distress.

3. Contraction stress test (CST)

CST evaluates fetal tolerance of contractions by assessing for late decelerations. **Negative** = no late decelerations with adequate contractions (reassuring). **Positive** = late decelerations with at least half the contractions (concerning). The test may be inappropriate when contractions could be unsafe, such as certain placenta or preterm labor risks.

4. Biophysical profile (BPP)

Component	0 or 2 points	What it reflects
NST	Reactive fetal heart-rate response	Acute oxygenation/autonomic function.
Fetal breathing	Observed breathing movements	Neurologic function.
Gross movement	Observed body/limb movements	Neurologic function.
Fetal tone	Extension/flexion activity	Neurologic function.
Amniotic fluid	Adequate fluid measurement	Longer-term placental/renal status.

Interpretation: 8-10 usually reassuring with adequate fluid; 6 equivocal and requires context/reassessment; 4 or less concerning and needs prompt provider evaluation. A low fluid result can be important despite a favorable composite score.

5. Ultrasound, Doppler and amniotic fluid

Assessment	Findings / implications	Nursing action
Ultrasound	Dating, anatomy, growth, placenta and fetal presentation	Explain procedure and follow-up findings.
Umbilical artery Doppler	Abnormal resistance may suggest placental insufficiency	Coordinate surveillance for growth restriction.
Oligohydramnios	Low fluid; risk of cord compression, fetal renal problems or membrane rupture	Assess membrane status and fetal monitoring needs.

Assessment	Findings / implications	Nursing action
Polyhydramnios	Excess fluid; associated with diabetes, anomalies and malpresentation	Monitor maternal symptoms and obstetric complications.

6. Genetic screening versus diagnosis

Test	Type	Nursing teaching
Cell-free DNA	Screening	Estimates likelihood of selected chromosomal conditions; positive screen needs counseling/confirmation.
Maternal serum AFP / quad	Screening	May identify increased risk of neural tube or chromosomal abnormalities.
CVS	Diagnostic placental sampling	Earlier invasive testing; review risks, indications and informed consent.
Amniocentesis	Diagnostic amniotic-fluid sampling	Review indications, procedural risks and postprocedure warning signs.
Cordocentesis/PUBS	Invasive fetal blood sampling	Specialized procedure; monitor for complications.

NCLEX PRIORITY: A nonreactive NST is not automatically fetal distress; a positive CST and low BPP are more concerning. Always interpret testing with gestational age and the full clinical picture.

MODULE 7 | CHAPTER 11 - PREEXISTING HEALTH CONDITIONS

1. Diabetes physiology in pregnancy

Placental hormones increase maternal insulin resistance as pregnancy progresses. Patients with pregestational diabetes need close monitoring before and during pregnancy because hyperglycemia early in organ formation raises congenital anomaly risk, while later poor control raises risks of excessive fetal growth, preeclampsia and neonatal hypoglycemia. Insulin needs vary by trimester and after delivery.

Hypoglycemia	Hyperglycemia / DKA	Immediate nursing reasoning
Glucose often below 70 mg/dL; sweating, hunger, tremor, confusion, pallor	Thirst, polyuria, dehydration, nausea, abdominal pain, deep/rapid breathing, ketones, altered LOC	Check bedside glucose; if alert and able to swallow, treat low glucose per protocol; DKA needs emergency fluids, insulin, electrolyte monitoring and maternal/fetal evaluation.

DKA can occur during pregnancy at lower glucose levels than in nonpregnant patients. Infection, vomiting and insulin omission can trigger it. Maternal stabilization and correction of acid-base/electrolyte abnormalities are urgent.

2. Maternal and fetal complications of diabetes

• Maternal: Hypoglycemia, DKA, worsening vascular/renal disease, hypertension and preeclampsia, infection, operative delivery.

• Fetal/neonatal: Congenital malformations (especially with early poor control), macrosomia or growth restriction, birth injury, neonatal hypoglycemia, respiratory complications and stillbirth risk.

• Care: Glucose log review, individualized nutrition, medication administration, activity counseling, fetal surveillance, and postpartum glucose follow-up.

3. Thyroid disease

Hypothyroidism: Fatigue, cold intolerance, constipation, slowed metabolism; untreated disease can impair maternal and fetal outcomes. Monitor TSH and thyroid replacement as ordered. **Hyperthyroidism:** Tachycardia, heat intolerance, weight loss, tremor; severe disease may precipitate thyroid storm. Monitor symptoms, labs and medication effects. Distinguish normal pregnancy changes from disease progression.

4. Cardiovascular, respiratory and renal disorders

Condition	Maternal risk	Priority assessment
Cardiac disease	Pregnancy increases circulating volume and cardiac workload	Dyspnea at rest, chest pain, hypoxemia, new crackles, worsening edema.
Asthma	Poor control can reduce maternal and fetal oxygenation	Breath sounds, respiratory effort, saturation and inhaler use.
Renal disease	Hypertension, fluid and electrolyte disturbances, preeclampsia risk	Creatinine, urine protein, I&O;, BP and edema.
Autoimmune disease / lupus	Flare, thrombosis, placental insufficiency	Symptoms, medications, fetal growth and laboratory monitoring.
Seizure disorder	Maternal trauma/hypoxia and medication concerns	Adherence, seizure frequency, safety and coordinated medication review.

5. Substance use disorder and psychosocial nursing

Screen without judgment; consider alcohol, nicotine, cannabis, opioids, stimulants and polysubstance use. Risks vary by substance and exposure. Encourage evidence-based treatment, coordinate prenatal and behavioral care, assess withdrawal and safety, and avoid punitive language. Patient education should be specific, supportive and realistic.

NCLEX PRIORITY: When a pregnant patient has hypoxia, DKA, seizure or cardiac decompensation, stabilize maternal physiology first; fetal oxygen delivery depends on maternal status.

MODULE 7 | CHAPTER 12 - GESTATIONAL CONDITIONS

1. Hypertensive disorder classification

Diagnosis	Diagnostic pattern	Clinical significance
Gestational hypertension	New BP at least 140/90 after 20 weeks without preeclampsia findings	Can progress to preeclampsia.
Preeclampsia	Hypertension plus proteinuria OR qualifying organ dysfunction after 20 weeks	Multi-organ endothelial dysfunction.
Severe-range hypertension	Systolic at least 160 OR diastolic at least 110	Urgent treatment; do not wait four hours to address sustained severe BP.
Eclampsia	Seizure attributable to preeclampsia	Airway and injury emergency.
Chronic hypertension	Predates pregnancy or occurs before 20 weeks	Risk of superimposed preeclampsia.
HELLP	Hemolysis, elevated liver enzymes, low platelets	May occur with only mild BP elevation or without proteinuria.

2. Why preeclampsia damages organs

Abnormal placental vascular remodeling and reduced placental perfusion contribute to widespread endothelial dysfunction and vasospasm. This raises BP but also reduces effective tissue perfusion. Capillary leak produces edema and reduced intravascular volume. Renal injury leads to proteinuria and reduced urine output; liver ischemia causes elevated enzymes and RUQ pain; cerebral irritability causes headache, hyperreflexia, clonus and seizures; pulmonary capillary leak can cause edema.

3. Diagnostic findings and warning signs

Finding	Interpretation
BP at least 140/90 on appropriately spaced measurements	Hypertension after 20 weeks warrants evaluation.
BP at least 160 systolic or 110 diastolic	Severe range: urgent response.
Protein at least 300 mg/24 h OR protein/creatinine at least 0.3	Proteinuria criterion when present.
Platelets below 100,000/mm3	Severe feature / possible HELLP.
Creatinine above 1.1 mg/dL or doubled from baseline	Renal involvement.
AST/ALT at least twice normal; RUQ pain	Hepatic involvement.
Persistent headache, visual symptoms, pulmonary edema	Severe organ dysfunction even without proteinuria.

Edema alone does not establish preeclampsia. Proteinuria quantity alone no longer determines severe features. HELLP is primarily a laboratory diagnosis and can present with nausea, malaise or RUQ pain.

4. Assessment, prevention and management

• Risk factors include prior preeclampsia, multifetal gestation, chronic hypertension, diabetes, kidney disease, lupus, obesity and first pregnancy.

• Low-dose aspirin 81 mg/day is recommended for selected high-risk pregnancies under obstetric direction, typically started between 12 and 28 weeks (ideally before 16 weeks).

• Monitor BP trends, headache, visual symptoms, RUQ pain, breath sounds, oxygenation, DTRs/clonus, I&O;, platelet count, liver enzymes and creatinine.

• Assess fetus with movement counts, NST/BPP, ultrasound growth and fluid studies as ordered.

• Severe disease requires hospitalization, seizure prophylaxis and antihypertensives when indicated; timing of birth balances maternal/fetal stability and gestational age.

• After delivery, continue monitoring because postpartum preeclampsia and seizures can occur; magnesium often continues approximately 24 hours when ordered.

5. Magnesium sulfate: medication safety

Aspect	What to remember
Purpose	Prevent or treat eclamptic seizures; it does NOT replace antihypertensive medication.
Typical IV regimen in chapter	Loading 4-6 g, then maintenance approximately 1-2 g/hour by pump per protocol.
Assess	Respiratory rate, patellar reflexes, level of consciousness, urine output, BP, I&O;, fetal status.
Report immediately	RR below 12/min, absent DTRs, urine output below about 25-30 mL/hr, marked sedation.
Toxicity action	Stop infusion, summon help, assess airway/respirations; prepare IV calcium gluconate as ordered.
Other considerations	High-alert drug; reduced renal excretion increases toxicity risk; use pump and verified concentration.

6. Antihypertensive medications

Medication	Mechanism / use	Monitor
Labetalol	Alpha/beta blocker used for severe BP and ongoing control	BP, pulse, asthma/bronchospasm and bradycardia risk.
Hydralazine	Arteriolar vasodilator	Rapid BP drop, tachycardia, headache and fetal perfusion.
Nifedipine	Calcium-channel blocker; oral treatment	BP, flushing, headache and hypotension.

Avoid lowering BP excessively or abruptly because uteroplacental perfusion may decline. Medication route, dose and timing follow current obstetric emergency protocols. Concurrent nifedipine and magnesium use may be clinically appropriate with monitoring; do not treat an older textbook caution as an absolute prohibition.

7. Eclamptic seizure: step-by-step nursing response

• Call for immediate assistance; remain with patient and note seizure onset/duration.

• Protect from injury, raise/pad side rails if feasible; do not restrain and do not place objects in mouth.

• Position laterally when possible to protect airway and reduce aspiration risk.

• Assess respirations, oxygenation, circulation and level of consciousness; prepare suction/oxygen/resuscitation equipment.

• Administer prescribed anticonvulsant treatment; reassess maternal and fetal status after seizure.

8. Hyperemesis gravidarum

Severe, persistent pregnancy vomiting can cause dehydration, weight loss, electrolyte disturbances and nutritional deficiency. Assess orthostatic symptoms, intake/output, weight, urine concentration/ketones when indicated and electrolyte labs. Management may require IV fluids, antiemetics, nutrition support and thiamine before dextrose in prolonged malnutrition. Differentiate from routine morning sickness by severity and consequences.

9. Bleeding and pregnancy loss: recognize patterns

Condition	Classic presentation	Nursing priority
Ectopic pregnancy	Early pain, spotting; shoulder pain, syncope or shock if ruptured	IV access, hemodynamic monitoring, emergency escalation.
Miscarriage	Bleeding/cramping with varying cervical findings	Assess stability, infection, ultrasound/hCG and emotional support.
Placenta previa	Often painless bright-red bleeding later in pregnancy	Avoid digital vaginal exam until excluded; assess blood loss/fetus.
Placental abruption	Painful bleeding, tender/rigid uterus, fetal compromise; concealed bleeding possible	Maternal resuscitation, continuous fetal assessment, rapid team response.

10. UTI, surgery and trauma in pregnancy

UTIs may progress to pyelonephritis; fever, flank pain, vomiting and systemic symptoms require prompt evaluation. Surgery requires coordinated obstetric and anesthesia care, maternal oxygenation and fetal considerations. With trauma, **maternal stabilization always comes first**; then assess fetal well-being, contractions, bleeding and possible placental abruption.

11. Postpartum follow-up and long-term risk

Postpartum hypertension and preeclampsia can present after hospital discharge. Teach urgent return for severe headache, visual symptoms, dyspnea, chest pain or upper abdominal pain. A history of preeclampsia increases future cardiovascular and chronic hypertension risk; communicate this history during future primary care.

APPENDIX | QUICK DIFFERENTIATION TABLES

Which patient needs attention first?

Presentation	Likely concern	First nursing priority
35 weeks, BP 170/112 and headache	Preeclampsia with severe features	Urgent evaluation, seizure precautions and BP treatment pathway.
On magnesium: RR 10, absent DTRs	Magnesium toxicity	Stop infusion, summon help, support airway, prepare antidote.
Early pregnancy, unilateral pain and syncope	Ruptured ectopic pregnancy	Treat as hemorrhagic shock emergency.
Third trimester, painless bright red bleeding	Placenta previa	No digital vaginal exam; maternal/fetal assessment.
Third trimester, rigid tender uterus and pain	Placental abruption	Resuscitation and immediate obstetric response.
DKA symptoms with pregnancy	Maternal metabolic emergency	Airway/circulation, fluids, insulin/electrolytes per protocol.
Decreased fetal movement	Possible fetal compromise	Prompt fetal evaluation.
After birth, severe headache/vision changes	Postpartum preeclampsia	Urgent evaluation.

Lab and threshold memorization

Value	Interpretation
Fetal baseline heart rate: 110-160 bpm	Usual baseline range.
Gestational hypertension: BP at least 140 systolic or 90 diastolic	Confirm appropriately; evaluate after 20 weeks.
Severe hypertension: BP at least 160 systolic OR 110 diastolic	Urgent treatment pathway.
Proteinuria: at least 300 mg/24 h or ratio at least 0.3	Supports preeclampsia diagnosis.
Platelets below 100,000/mm3	Severe feature / HELLP consideration.
Creatinine above 1.1 mg/dL or doubled	Renal severe feature.
Magnesium monitoring: RR below 12; UO below 25-30 mL/hr; absent DTRs	Toxicity concerns; follow protocol.
Reactive NST at 32+ weeks: 2 accelerations, 15x15 in 20 min	Reassuring surveillance result.

Value	Interpretation
BPP: 8-10 reassuring, 6 equivocal, 4 or less concerning	Interpret with amniotic fluid and clinical context.

PRACTICE EXAM | 20 APPLICATION QUESTIONS

Answers are intentionally omitted. Record your choices and send them for grading with rationales. SATA means select all that apply.

1. A 32-week patient has BP 166/108, persistent headache and visual spots. What should the nurse do first?

A. Schedule a visit in one week B. Escalate for urgent maternal evaluation C. Encourage oral fluids D. Reassure that edema is normal

Answer: _____

2. Which laboratory finding supports HELLP syndrome?

A. Platelets 245,000 B. AST within normal limits C. Platelets 78,000 with elevated AST D. Hemoglobin 13.5 only

Answer: _____

3. The patient on magnesium has RR 10/min and no patellar reflexes. Priority?

A. Continue at same rate B. Stop infusion and call for help C. Give extra loading dose D. Encourage walking

Answer: _____

4. Which statement about magnesium sulfate is correct?

A. It is the primary antihypertensive B. It prevents/treats seizures C. It cures HELLP D. It increases urine output

Answer: _____

5. A patient at 34 weeks has painless vaginal bleeding. Which action should be avoided until previa is excluded?

A. BP monitoring B. Fetal monitoring C. Digital vaginal examination D. IV access

Answer: _____

6. A patient at 37 weeks has a rigid, tender uterus and vaginal bleeding. Most likely?

A. Placental abruption B. Uncomplicated labor C. Candidiasis D. Placenta previa

Answer: _____

7. A reactive NST at 34 weeks generally requires which finding?

A. Two 15x15 accelerations B. Late decelerations C. No fetal movement D. Three contractions

Answer: _____

8. Which CST result is reassuring?

A. Positive B. Negative C. Recurrent late decelerations D. Variable decelerations with every contraction

Answer: _____

9. A BPP score of 4/10 suggests what action?

A. Routine annual follow-up B. Prompt further evaluation C. Ignore if patient feels well D. Discharge without instructions

Answer: _____

10. Which is a positive sign of pregnancy?

A. Amenorrhea B. Breast tenderness C. Fetal cardiac activity on ultrasound D. Positive home pregnancy test

Answer: _____

11. A pregnant patient has fatigue, pallor and pica. Which problem should the nurse investigate?

A. Iron deficiency B. Placenta previa C. Gestational hypertension only D. Normal fetal movement

Answer: _____

12. Which food choice is most concerning in pregnancy?

A. Fully cooked poultry B. Pasteurized yogurt C. Unpasteurized soft cheese D. Washed fruit

Answer: _____

13. Which patient is at risk for magnesium toxicity?

A. Adequate urine output B. Declining renal function C. Normal reflexes D. Normal respiratory rate

Answer: _____

14. A pregnant patient with diabetes has abdominal pain, vomiting, rapid breathing and ketones. Priority?

A. Reassure about morning sickness B. Evaluate urgently for DKA C. Give an oral snack only D. Wait for the next appointment

Answer: _____

15. Which contraception offers STI risk reduction?

A. Implant B. Hormonal IUD C. Condom D. Combined pill

Answer: _____

16. Which finding warrants urgent assessment after delivery?

A. Mild fatigue B. Severe headache and blurred vision C. Normal lochia D. Mild afterpains

Answer: _____

17. A patient with suspected PID asks why treatment is urgent. Best explanation?

A. Untreated infection can harm fertility B. It always resolves untreated C. It is never infectious D. Antibiotics are never used

Answer: _____

18. A patient after ovarian stimulation has rapid weight gain, abdominal distension and low urine output. Concern?

A. Expected ovulation pain only B. Ovarian hyperstimulation syndrome C. Menopause D. Normal menses

Answer: _____

19. A patient with a positive cell-free DNA screen asks whether the fetus definitely has a chromosome condition. Best reply?

A. Yes, screening is diagnostic B. No; discuss diagnostic confirmation options C. Ignore result D. Repeat only after birth

Answer: _____

20. In a trauma involving a pregnant patient, what is the first priority?

A. Fetal ultrasound before anything else B. Maternal airway/breathing/circulation C. Prenatal vitamin review D. Routine discharge teaching

Answer: _____

SOURCE AND STUDY NOTES

Prepared from the nine user-supplied chapter text uploads, October 8, 2026: Giddens Chapter 19 Reproduction; Giddens Chapter 20 Sexuality; Women's Reproductive Health and Gynecologic Disorders; Infertility, Assisted Reproductive Technologies, and Contraception; Chapters 8-12 on prenatal care, nutrition and high-risk pregnancy.

The source chapters include some older recommendations and medication cautions. For clinical practice, follow your instructor's assigned standards, current professional guidance, provider orders and facility protocols. This guide emphasizes core principles and avoids treating historical drug doses or dated practices as universal current instructions.

Suggested study method: for each chapter, make a one-minute verbal explanation of pathophysiology; identify three red flags; name the key diagnostic tests; explain what the nurse monitors after the main medications; then complete the questions without looking at notes.